**9th Class • Punjab Board • PECTAA New Syllabus**

# Chapter 7: Emerging Technologies in Computer Science

Complete solved exercise — artificial intelligence, machine learning, IoT, applications and security.

**Chapter 7 • Complete official exercise • 6 MCQs + 6 short + 3 long questions**

### What this chapter is about

• **Artificial Intelligence (AI)** enables machines to perform tasks associated with human intelligence, such as learning, understanding language, recognising images and making decisions.

• **Machine learning** learns patterns from data. White-box models expose understandable reasons; black-box models may be accurate but difficult to explain.

• The **Internet of Things (IoT)** connects physical devices through sensors, networks, processing platforms and applications so they can collect data and act on it.

• AI and IoT offer healthcare, education and transport benefits, but require privacy, security, fairness, reliability and responsible human oversight.

**Coverage:** Chapter 7 corresponds to Unit 10 Emerging Technologies in Computer Science in the twelve-unit textbook. The April 2026 nine-unit exercise is solved in book order.

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## A. Multiple Choice Questions

**1. Which of the following is not a subfield of AI?**

(a) Machine Learning (b) Natural Language Processing
(c) Computer Vision (d) DBMS

**Answer: (d) DBMS** — DBMS means Database Management System; it manages data and is not itself an AI subfield.

**2. Which of these is a security concern in IoT deployments?**

(a) Device vulnerability (b) Data privacy
(c) Lack of standardization (d) All of the above

**Answer: (d) All of the above** — The textbook treats vulnerable devices, privacy risks and inconsistent standards as linked IoT security/deployment concerns.

**3. Which of the following is an application of AI in healthcare?**

(a) Personalized drug development (b) Automated diagnosis
(c) Remote patient monitoring (d) All of the above

**Answer: (d) All of the above** — AI can support drug discovery, analyse diagnostic data and monitor patient measurements, subject to clinical oversight.

**4. What is the primary purpose of using AI techniques in machine learning models?**

(a) To improve accuracy
(b) To enhance interpretability
(c) To reduce computational complexity
(d) All of the above

**Answer: (a) To improve accuracy** — The primary aim is to learn useful patterns that improve prediction or decision accuracy; interpretability and efficiency are separate desirable goals.

**5. What is the key difference between explainable (whitebox) and unexplainable (blackbox) AI models?**

(a) The complexity of the model
(b) The ability to understand the decision-making process
(c) The performance of the model
(d) The training data used

**Answer: (b) The ability to understand the decision-making process** — White-box models let people inspect reasons for a decision; black-box models do not provide an easily understandable internal decision process.

**6. Which of the following is an application of IoT in the transportation domain?**

(a) Smart traffic management
(b) Vehicle-to-Vehicle (V2V) communication
(c) Predictive maintenance of vehicles
(d) All of the above

**Answer: (d) All of the above** — Connected transport sensors support traffic control, vehicle communication and early detection of maintenance needs.

## B. Short Questions

**1. Define Artificial Intelligence (AI).**

Artificial Intelligence is the branch of computer science that develops systems capable of tasks normally associated with human intelligence, such as learning from experience, understanding language, recognising objects, solving problems and making decisions.

**2. Provide two examples of AI applications in healthcare.**

Two examples are **automated diagnostic support**, where AI analyses scans or patient data to flag possible disease, and **remote patient monitoring**, where AI examines wearable-device measurements and alerts staff to unusual changes. Doctors must verify important clinical decisions.

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### 3. Explain the role of AI techniques in advancing machine learning models.

AI techniques help models learn useful patterns, optimise their parameters and improve the accuracy of predictions or classifications. Better feature learning supports language, vision and recommendation tasks, while evaluation and explainability techniques help measure performance and understand decisions.

### 4. Define the Internet of Things (IoT).

The Internet of Things is a network of physical objects fitted with sensors, software and communication capability that allows them to collect and exchange data and, in many cases, perform actions through actuators with limited human intervention.

### 5. Describe the significance of IoT in connecting devices and systems.

IoT links physical equipment with digital platforms, enabling real-time monitoring, remote control, automation and data-based decisions. Connected devices can coordinate across a home, hospital, vehicle or factory, improving efficiency and response time when security and interoperability are managed properly.

### 6. Describe applications of IoT in Transportatin Domain.

IoT supports **smart traffic management** through road sensors and adaptive signals, **vehicle-to-vehicle communication** for safety warnings, **fleet tracking** for location and fuel monitoring, and **predictive maintenance** through engine/vehicle sensors that detect faults before breakdown.

## C. Long Questions

### 1. Discuss the various applications of AI in the field of education.

AI can support teachers and learners by analysing educational data and automating suitable parts of feedback, practice and administration. It should assist—not replace—professional judgement and human interaction.

• **Personalised learning:** A system can adjust lesson difficulty, practice and pace according to a learner's previous answers, giving extra support where needed.

• **Intelligent tutoring:** AI tutors can answer routine questions, give hints and guide students through step-by-step exercises at any time.

• **Assessment and feedback:** Objective questions can be graded automatically, and writing or coding tools can highlight areas for improvement before a teacher reviews the work.

• **Early support:** Attendance, participation and assessment patterns may identify students at risk of falling behind so that teachers can offer timely help.

• **Accessibility:** Speech-to-text, text-to-speech, translation, captioning and image description can make learning material easier to access for different needs and languages.

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• **Content and lesson planning:** Teachers can use AI to draft quizzes, examples and differentiated resources, then check accuracy, level and curriculum fit before use.

• **Administration:** AI can assist with timetabling, routine enquiries and organising records, giving staff more time for teaching.

### Responsible educational use

• Protect student data, obtain appropriate consent and avoid uploading identifiable records to unapproved services.

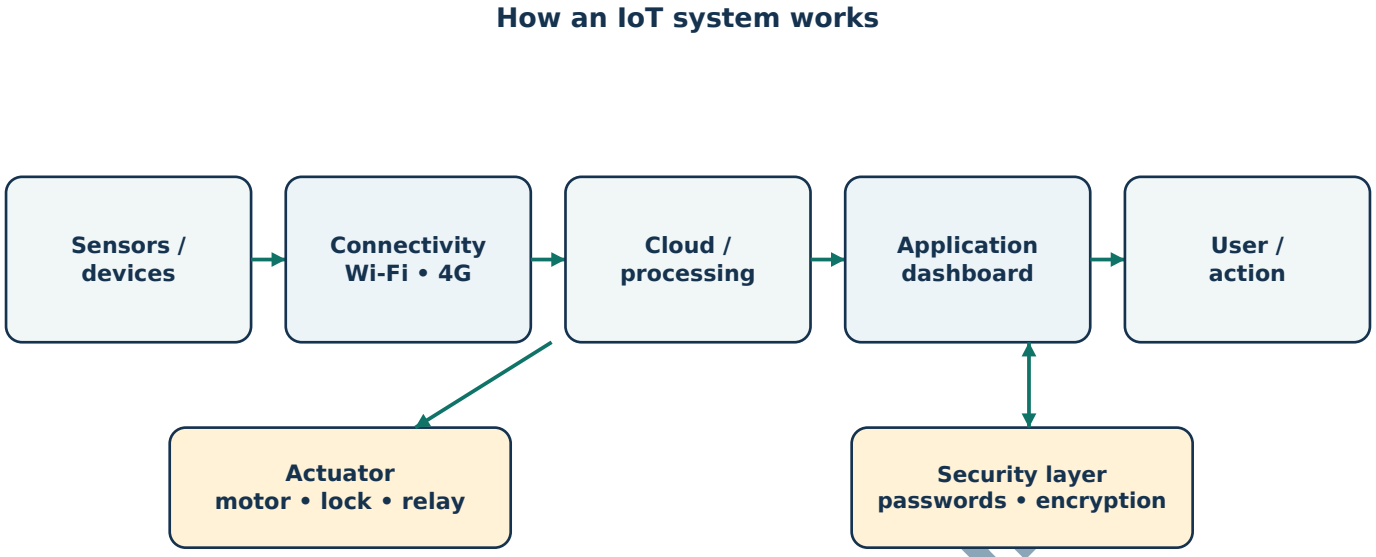
• Check bias, incorrect answers and inappropriate content; AI-generated work needs teacher verification.

• Keep assessment fair and transparent, preserve academic integrity and ensure that final high-stakes decisions include accountable human review.

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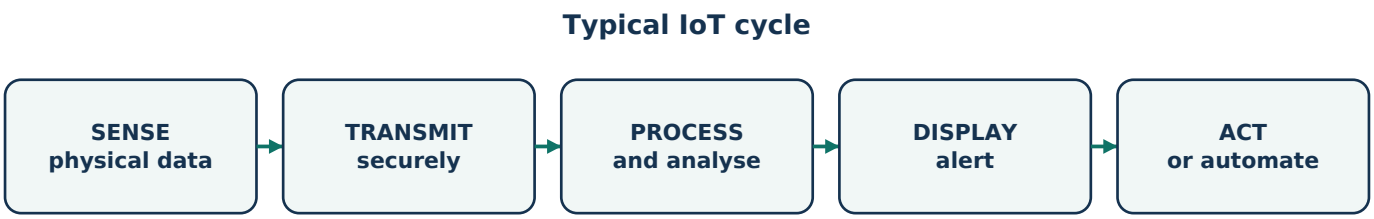
## 2. Describe the components of an IoT system.

**Figure 1 — Main IoT components and information flow**




| Component | Role | Example |
|---|---|---|
| Sensors / devices | Measure physical conditions or collect identifiers. | Temperature, motion, GPS, heart-rate or vehicle sensors. |
| Actuators | Convert a digital command into a physical action. | Motor, relay, smart lock, valve or alarm. |
| Connectivity / gateway | Carry data between devices and platforms; a gateway may translate protocols and filter data locally. | Wi-Fi, Bluetooth, 4G/5G, Ethernet or an IoT gateway. |
| Data processing / cloud | Store, clean and analyse data, apply rules and sometimes run AI models. | Cloud platform or edge computer detecting overheating. |
| Application / dashboard | Present readings, alerts and controls to users and other systems. | Mobile transport dashboard showing vehicles on a map. |
| User / business process | Interpret information, configure rules and take responsibility for action. | Operator schedules maintenance after an alert. |
| Security and management | Authenticate devices/users, encrypt communication, update firmware and control permissions. | Unique credentials, TLS, secure updates and logs. |

**Figure 2 — Sense-communicate-process-act**




In a smart vehicle example, an engine sensor collects temperature, a gateway transmits it, the platform detects an abnormal value, a dashboard warns the operator and maintenance is arranged. If an actuator is used, a command travels back through the system to perform the action.

### 3. Write a note on sub-fields of AI.

| AI subfield | What it does | Example application |
|---|---|---|
| Machine Learning (ML) | Learns patterns and rules from data instead of relying only on manually written rules. | Predicting equipment failure or classifying email as spam. |
| Deep Learning | Uses multi-layer neural networks to learn complex features, especially from large datasets. | Image recognition and advanced language models. |
| Natural Language Processing (NLP) | Works with human spoken or written language. | Translation, text classification, chat assistants and summarisation. |
| Computer Vision | Interprets images and video. | Reading number plates or assisting analysis of medical scans. |
| Speech / audio intelligence | Recognises and generates speech and analyses sound. | Voice assistants, transcription and machine-fault sound detection. |
| Robotics | Combines sensing, decision-making and physical action. | Warehouse robots and automated inspection. |
| Expert / knowledge-based systems | Represents specialist knowledge and applies rules to reason about problems. | Rule-based troubleshooting or decision support. |

These fields overlap. A medical robot may use computer vision to inspect an image, machine learning to classify it, NLP to read a report and robotic control to perform a physical task. The system's usefulness still depends on suitable data, testing and safe human supervision.

## D. More Important Questions (from the text)

### 1. Compare white-box and black-box AI models.

| White-box / explainable model | Black-box / difficult-to-explain model |
|---|---|
| People can inspect the rules, features or reasoning behind an output. | The internal relationship between input and output is difficult for people to interpret directly. |
| Often simpler structures such as a small decision tree or clear rule set. | Often large, highly interconnected models, although complexity alone is not the defining feature. |
| Easier to audit, justify, debug and check for inappropriate rules. | May perform very well but needs additional testing and explanation methods to build justified trust. |
| May be less flexible for some complex patterns. | Can learn complex patterns but may require more data and computation. |
| Useful where explanations are legally, medically or socially important. | Requires careful monitoring for bias, errors and unexpected behaviour. |

### 2. State major security and privacy considerations in IoT deployments.

• Change default passwords and give every device and user unique, least-privilege credentials.

• Encrypt data during transmission and storage and protect encryption keys.

• Install signed security updates and replace devices that no longer receive safe firmware support.

• Isolate IoT devices on suitable network segments and disable unused services and ports.

• Collect only necessary personal data, define retention, obtain consent where needed and control sharing.

• Check interoperability and standards so that devices from different suppliers can be managed securely.

• Monitor logs and behaviour for unusual activity, and keep an incident-response and safe manual-fallback plan.

### 3. Give a brief outline of the development of AI.

Early work asked whether machines could show intelligent behaviour; the term **artificial intelligence** became established in the 1950s. Early systems relied heavily on symbolic rules and problem-solving programs. Later expert systems represented specialist knowledge but were limited by the effort required to write and maintain rules.

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Machine learning then shifted attention towards learning statistical patterns from data. Larger datasets, faster processors and improved neural-network methods enabled deep learning, producing major advances in vision, speech and language. Modern generative systems can create text, images and other content, but their fluency does not guarantee truth or understanding.

Throughout this development, the central lesson is unchanged: evaluate real performance and limitations, protect people and data, and use human oversight appropriate to the risk.

## Points to Remember

**AI:** computer systems performing tasks associated with learning, language, perception, reasoning and decisions.

**ML:** learns patterns from data; deep learning uses multi-layer neural networks.

**White-box:** understandable decision process. **Black-box:** difficult-to-interpret decision process; accuracy alone is not enough for trust.

**AI subfields:** ML, NLP, computer vision, robotics, speech and expert/knowledge-based systems; DBMS is not one.

**IoT:** sensors/devices + connectivity + processing + application + users; actuators perform physical actions.

**Transport uses:** smart traffic, V2V warnings, fleet tracking and predictive maintenance.

**Education uses:** personalised practice, tutoring, feedback, accessibility and administration with teacher oversight.

**Safe deployment:** privacy, fairness, unique credentials, encryption, updates, monitoring and accountable human review.

### MCQ key at a glance

| Q | Key | Q | Key | Q | Key | Q | Key | Q | Key |
|---|---|---|---|---|---|---|---|---|---|
| 1 | d | 2 | d | 3 | d | 4 | a | 5 | b |
| 6 | d | | | | | | | | |

### How to attempt this chapter in the paper

- For application questions, give the use, the data involved and the benefit; add one limitation where human safety or fairness matters.
- For an IoT-components answer, show both directions: sensor data travels to the platform, while commands can return to actuators.
- Do not call DBMS an AI subfield; it stores/manages data that an AI system may use.
- For white-box versus black-box, focus on interpretability of the decision process rather than simply model size or accuracy.

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